

MORTIMER: Multi-asset Optimisation and Risk Targeting with Integrated Monthly E6 Rebalancing

Nelson Alvarez, Amir Hajomar, Sean Majewski, and Miles Rack

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Summary

MORTIMER combines monthly equal-risk-contribution (ERC) weights with a causal volatility overlay across six exchange-traded funds (ETFs). In the held-out out-of-sample (OOS) test, the portfolio returned 9.52% annualised after five basis points of trading costs, with 7.47% volatility, a 0.714 Sharpe ratio, a -7.00% maximum drawdown, and annual half-L1 turnover of 1.119. Relative to matched unscaled ERC, volatility fell by 8.2% and drawdown by 9.9%, while the Sharpe ratio also declined. Validation showed larger risk reductions alongside a 15-basis-point annual return difference. The results indicate that volatility targeting moderated portfolio risk without improving risk-adjusted return.

Economic hypothesis

Investors accept equity, credit, duration, and commodity risk premia, while benchmark mandates and slow risk-budget adjustment can leave portfolios exposed as volatility rises. Absolute and squared returns exhibit persistent dependence across financial markets, the empirical basis for treating recent variance as informative about near-term risk (Cont, 2001; Engle, 1982; Bollerslev, 1986). Realized-volatility research applies this persistence to forecasting future risk (Andersen et al., 2003), while volatility-managed portfolios apply it to portfolio exposure (Moreira and Muir, 2017). MORTIMER reduces risky exposure when forecast volatility exceeds its target. We test whether this rule lowers volatility and tail losses after costs, while limiting the associated return drag. A shock can arrive before the overlay responds, and lower exposure can delay recovery.

Data and universe

The fixed universe comprises QQQ, IWM, HYG, TLT, GLD, and DBC. Yahoo adjusted closes include distributions and splits; raw closes and volume support liquidity estimates (Yahoo Finance, n.d.). Cash accrues from initial-release observations in the Archival Federal Reserve Economic Data (ALFRED) DGS3MO series (Federal Reserve Bank of St. Louis, n.d.-a, n.d.-b), using the next exchange session for availability and actual/365 day-count accrual. Factor returns use the Kenneth French data library (French, n.d.). The New York Stock Exchange (NYSE) calendar spans 2 January 2008 to 2 October 2026, with 550 sessions reserved for estimation before scored returns. Training ends on 31 December 2018, and validation runs from 1 January 2019 to 1 October 2024. The held-out OOS period runs from 2 October 2024 to 2 October 2026, the shorter of the most recent 20% of sessions and two years. The six funds were selected for continuous price and volume histories through the end of the sample; membership is fixed throughout the analysis.

Methodology

At each month-end, estimate daily covariance from 126 completed observations using Ledoit–Wolf shrinkage toward a scaled-identity target (Ledoit and Wolf, 2004). The constrained equal-risk-contribution (ERC) allocation follows the risk-budgeting framework of Maillard, Roncalli, and

Teiletche (2010) and solves

$$\min_b \sum_{i=1}^6 \left[b_i (\hat{\Sigma}_t b)_i - \frac{b^\top \hat{\Sigma}_t b}{6} \right]^2, \quad \mathbf{1}^\top b = 1, \quad 0 \leq b_i \leq 0.35.$$

Here b_i is the base weight, $\hat{\Sigma}_t$ is daily return covariance, and $b_i(\hat{\Sigma}_t b)_i$ is variance contribution. Binding name caps can prevent exact equality. Shrinkage limits covariance estimation error, which can materially impair optimised allocations out of sample (DeMiguel, Garlappi, and Uppal, 2009).

For current base b_t , calculate historical portfolio returns $x_s = b_t^\top r_s$ using observations through close t . An exponentially weighted moving average (EWMA) with span n applies geometrically declining weights to squared returns. The annualised volatility forecast and capped risky exposure are

$$\hat{\sigma}_t = \sqrt{252 \max\{\text{EWMA}_{20}(x_s^2), \text{EWMA}_{63}(x_s^2)\}}, \quad k_t = \min(1, 0.10/\hat{\sigma}_t).$$

Each estimate requires 63 observations. Taking the larger estimate combines a faster response with a slower volatility measure. A five-percentage-point exposure change triggers a trade, while monthly base rebalances proceed independently. Gross exposure cannot exceed one, and the strategy neither shorts nor borrows. The 10% forecast target does not fix realised volatility.

The close- t signal is executed at close $t + 1$, after that session's return; new holdings first earn the next close-to-close return. Given pre-fee risky holdings h_i , portfolio net asset value (NAV) V , desired weights w_i , and cost rate c , post-fee NAV satisfies

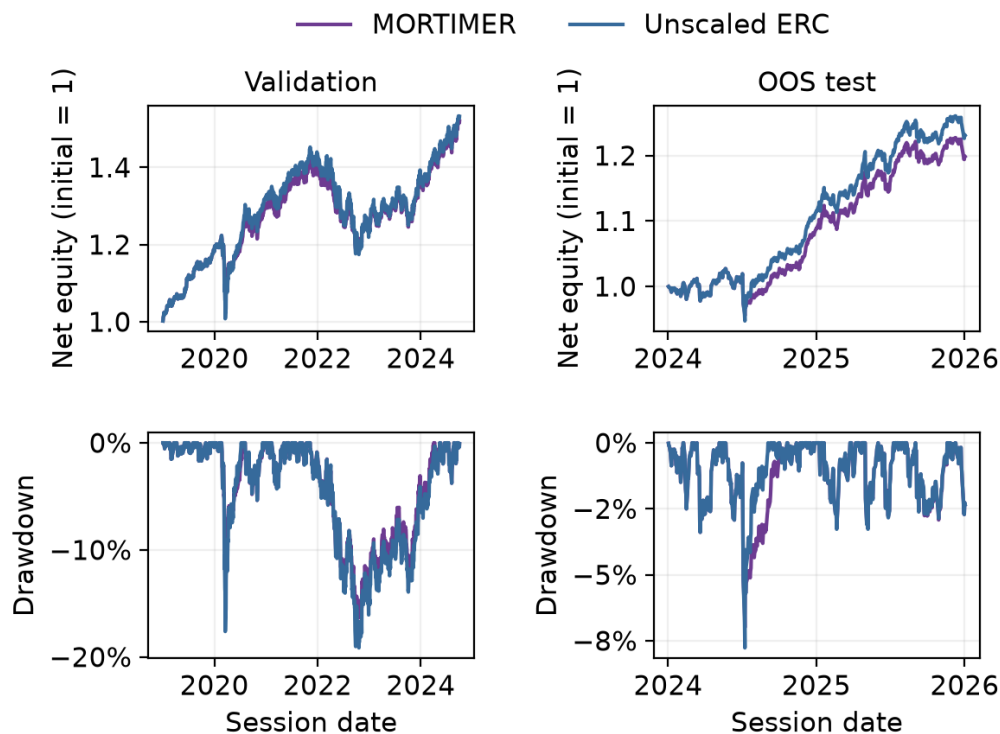
$$V^{post} = V - c \sum_i |V^{post} w_i - h_i|.$$

Costs are five basis points of risky dollars bought plus sold, representing spread, commission, and closing slippage; cash transfers carry no charge. A ten-basis-point cost case is evaluated on development data. The cash return proxy is derived from initial-release three-month Treasury bill yields.

Results

We calculate annualised return as compound annual growth rate (CAGR) from split-local net wealth using 252 trading sessions per year. Annualised volatility is the sample standard deviation of daily returns multiplied by $\sqrt{252}$, and the Sharpe ratio uses daily excess returns relative to the cash return proxy (Sharpe, 1994). Maximum drawdown is measured from initial wealth of one. Annual turnover is half the L1 change in risky and cash weights, and transaction costs apply to risky purchases and sales. The table reports all metrics net of five-basis-point costs for each split. Net equity begins at one in each split; the Results figure plots equity and drawdown separately for validation and the held-out test. A separate figure traces rolling volatility, risky exposure, and trailing annual turnover over the full sample, with dashed lines marking the training, validation, and test boundaries.

Sample	Rule	Ann. return%	Vol.%	Sharpe	Sortino	MDD%	Calmar	Turnover
Training	MORTIMER	5.85	6.37	0.859	1.232	-12.36	0.473	0.96
Training	Unscaled ERC	6.13	6.51	0.884	1.273	-12.48	0.491	0.45
Validation	MORTIMER	7.56	7.93	0.660	0.916	-16.63	0.455	1.57
Validation	Unscaled ERC	7.71	9.78	0.566	0.781	-19.14	0.403	0.35
OOS test	MORTIMER	9.52	7.47	0.714	0.984	-7.00	1.360	1.12
OOS test	Unscaled ERC	11.00	8.13	0.827	1.181	-7.77	1.416	0.35

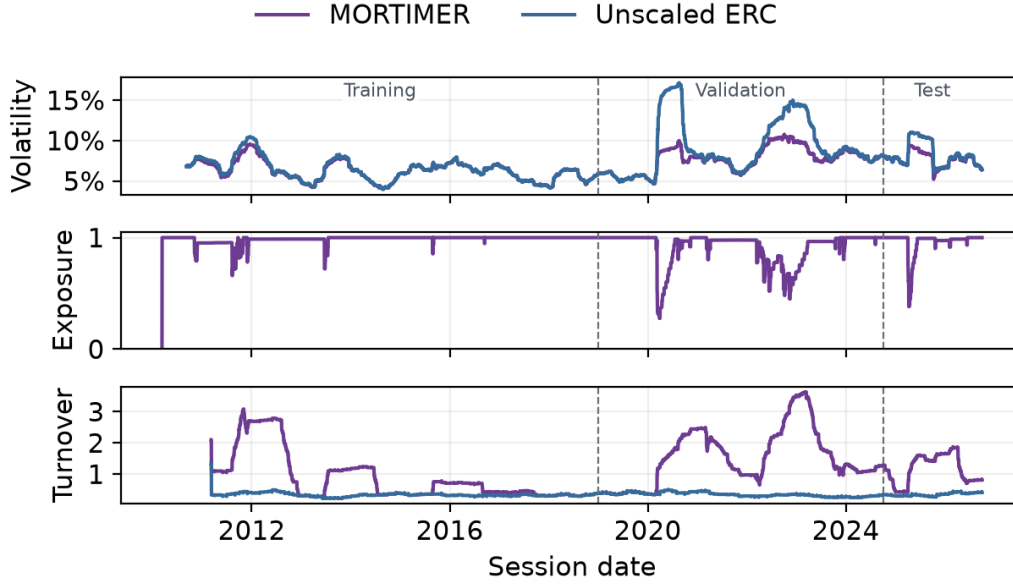


The volatility overlay has its clearest effect in validation. Relative to unscaled ERC, annualised volatility falls from 9.78% to 7.93%, and maximum drawdown from -19.14% to -16.63%; annualised return is lower by 15 basis points. At ten basis points per trade, validation CAGR and Sharpe are 7.46% and 0.649. A paired 20-session block bootstrap estimates a 95% interval of [-2.94, 2.44] percentage points for the validation return difference. In the held-out OOS test, volatility and maximum drawdown are lower by 8.2% and 9.9%, respectively; the Sharpe ratio is 0.714, compared with 0.827 for unscaled ERC. The forecast’s validation Pearson and rank information coefficients are 0.537 and 0.668 against following-window realised volatility.

Risk management

Base weights are long-only and capped at 35% per instrument. Gross exposure cannot exceed 100%, with the remainder held as cash. Weights drift between executions and may exceed target caps. The overlay reduces or restores exposure when the forecast changes by five percentage points.

Ordinary least squares (OLS) attribution on 479 common sessions estimates betas of 0.272 to the market, 0.031 to value, 0.066 to momentum, and 0.198 to Treasury-duration returns.



Liquidity and capacity

Liquidity is measured with previous-session trailing 20-session average daily dollar volume (ADDV). Participation is $Q/ADDV$, where Q is risky trade notional. The square-root impact scenario follows the empirical market-impact relation documented by Tóth et al. (2011):

$$I = Y\sigma\sqrt{Q/ADDV}, \quad \text{NAV cost} = IQ/NAV,$$

We set $Y = 0.5$ and use lagged volatility σ for each AUM scenario.

Initial AUM (\$m)	Max $Q/ADDV\%$	Impact-net CAGR%
1	0.52	7.55
10	5.22	7.53
100	52.16	7.47
500	260.81	7.35
1000	521.62	7.26

At $Y = 0.5$, a 1% participation limit is reached at initial AUM of \$1.92 million. The higher AUM rows exceed 100% participation and show the impact formula extrapolated beyond that level.

Limitations and further work

The fixed universe omits discontinued funds. Point-in-time membership and transaction-level execution data would support survivorship adjustment and cost estimates across the AUM range.

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